Process Synchronization using xv6-public

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1. Setup and kernel support

The x86 xv6-public implementation adds sync.c with shmget(), kernel sleep/wakeup semaphores, and a VM deallocation guard for SHMBASE. Each process attaches to the same kernel-owned page after fork(), because ordinary xv6 fork() copies the user address space. The tested macOS environment uses i686-elf-gcc and qemu-system-i386.

2. Peterson's algorithm

peterson.c uses shared flag[2] and turn variables for mutual exclusion without an OS lock. Ten critical-section entries per process produce a final counter of 20, with 0 mutual-exclusion violations. Result: PETERSON PASS.

3. Producer-consumer

prodcons.c uses a five-slot circular buffer with empty=5, full=0, and mutex=1. The producer inserts values 1 through 20 and the consumer removes them FIFO. Result: 20 produced, 20 consumed, PRODCONS PASS.

4. Readers-writers

readwrite.c creates three readers and two writers. RLOCK protects read_count, RESOURCE protects the shared value, and TURNSTILE prevents continuous new readers from postponing writers. Multiple readers overlap, writers are exclusive, final data is 8, violations are 0. Result: READWRITE PASS.

5. Dining philosophers

dining.c creates five philosophers and five fork semaphores. Resource ordering always acquires the lower-numbered fork before the higher-numbered fork, eliminating circular wait. All philosophers complete 5/5 cycles without a permanent hang. Result: DINING PASS.

6. Conclusion

The four programs demonstrate user-level Peterson mutual exclusion, kernel-assisted bounded-buffer synchronization, concurrent readers with exclusive writers, and deadlock-free resource ordering. QEMU logs and screenshots provide reproducible evidence for each question.

Question	Technique	Verification
Peterson	Flags + turn	Counter 20; violations 0
Producer-consumer	empty/full/mutex	20 produced; 20 consumed FIFO
Readers-writers	count + resource + turnstile	Final data 8; violations 0
Dining	Ordered fork acquisition	All philosophers complete 5 cycles